Kristu Jayanti College
AUTONOMOUS Bengaluru
Reaccredited A++ Grade by NAAC | Affiliated to Bengaluru North University

WEATHER APP

Project Report submitted in partial fulfilment of the requirements

for the Award of the Degree of

BACHELOR OF COMPUTER APPLICATIONS

Submitted By

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2024 – 2025

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CERTIFICATE FROM MENTOR

This is to certify that this internship work titled **WEATHER APP** is based on an original project study conducted by **Sabeena Kachary**, (22BCAB54) of VI semester BCA B under my mentorship.

This project work has not formed the basis for the award of any degree/ diploma by Bangalore North University or any other university.

Place:

Date:

Dr. GOPINATH D

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CERTIFICATE FROM DEPARTMENT

This is to certify that this project work titled **WEATHER APP** is based on an original project study conducted by **Sabeena Kachary** ,(22BCAB54) of VI semester BCA B under the mentorship Dr. Gopinath D.

This project work is based on the original and has not formed the basis for the award of any degree/diploma by Bangalore North University or any other University.

Head of the Department

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DECLARATION FROM CANDIDATE

I, **Sabeena Kachary** hereby declare that this project work titled **Weather App** is based on the original project conducted by me under the guidance of Dr. Gopinath D.

This has not been submitted earlier for the award of any other degree/diploma from Bangalore North University or any other University.

Place: Bengaluru

Date :

Roll No: 22BCAB54 Sabeena Kachary

Acknowledgement

"The success is the result of preparation, hard work, and learning from failure."

It's my duty to acknowledge and thank the individuals who has contributed to the successful completion of the project.

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1. Project Synopsis

The **Weather App** is a user-centric mobile application designed to deliver accurate and real-time weather information efficiently. Built using modern mobile development frameworks, this lightweight app integrates the OpenWeatherMap API. The app provides current weather conditions alongside a 5-day forecast for Bengaluru and other user-selected cities.

The application offers a simple yet powerful solution for users who wish to access weather updates and plan trips effectively. Upon launching the app, users can log in or sign up, with Bengaluru set as the default location, ensuring personalized access to weather data. Once authenticated, they are greeted with a clean and intuitive home screen where they can view current weather conditions, a 5-day forecast, or explore seasonal weather for travel destinations like Goa or Coorg.

The intuitive user interface allows users to easily check Bengaluru's weather, search for other cities, or save trip plans for future reference. Weather data and forecasts are presented in a clear, accessible format, with options to explore seasonal trends for travel planning.

The app follows clean architecture practices, leveraging cross-platform frameworks like Kotlin, offering smooth navigation and responsive UI. Unlike traditional weather apps, this application operates without a database, relying on local storage for user authentication (login/signup) and trip data persistence. This approach ensures a lightweight and efficient user experience, minimizing server dependency while maintaining functionality. While streamlined in its current form, the project lays the foundation for future enhancements like weather alerts, visual weather charts, multi-language support, and calendar integration.

Overall, this project showcases essential mobile development skills while delivering a functional and practical solution for weather tracking and trip planning, catering to travelers, weather enthusiasts, and anyone seeking reliable weather insights.

1.1 Purpose:

The “Weather App” is a streamlined yet powerful mobile application designed to provide users with accurate and real-time weather information. It allows users to securely log in or register via local storage and access weather updates for Bengaluru and other cities using the OpenWeatherMap API. The application offers a smooth and reliable user experience by providing a simple interface for checking current conditions, 5-day forecasts, and seasonal weather for trip planning, ensuring users can make informed decisions effortlessly.

By focusing on essential features, the app targets users who value simplicity and functionality without compromising on reliable weather data. It is designed to work efficiently even without an internet connection, offering robust offline access to saved trip plans and user preferences using local storage.

1.2 Goals:

The Weather App project is driven by three core goals to ensure functionality, usability, and personalization:

- **Goal 1: Deliver Accurate Weather Data**
  The app aims to provide precise, real-time weather information for Bengaluru and user-selected cities using the OpenWeatherMap API. This includes current conditions and temperature and 5-day forecasts, critical for daily planning or travel. Achieving this goal requires robust API integration, efficient request handling, and error management for network issues. The focus is on achieving 99% data accuracy and response times under 2 seconds, ensuring users trust the app for timely updates.

- **Goal 2: Ensure Seamless User Interaction**
  A user-friendly interface is central to the app’s success. This goal involves developing intuitive login/signup screens, a responsive search bar, and accessible buttons for forecasts and trip planning. Built in Android Studio using Kotlin, the app targets load times below 2 seconds and crash-free navigation. User testing will refine the design to achieve 95% satisfaction, ensuring effortless interaction for all users, from tech-savvy travelers to casual weather checkers.

- **Goal 3: Enable Personalized Trip Planning**
  The app seeks to enhance user engagement by allowing trip plans (city + season) to be saved and viewed later. For example, a user can save “Goa, Summer” to check its weather later.

Using local storage (SharedPreferences), this feature ensures data persistence without a database. The goal includes 100% retrieval accuracy for saved trips, catering to travelers planning seasonal getaways and adding a personalized layer to the weather experience.

1.3 Intended Audience:

The Weather App is designed for a diverse audience with specific weather-related needs:

- **Primary Audience**:
  - **Bengaluru Residents**: Individuals aged 18-60 seeking daily weather updates for commuting, outdoor activities, or event planning. They rely on accurate Bengaluru weather data and occasional checks for other cities.
  - **Travelers**: People planning trips to Indian cities like Goa, Delhi, or Mumbai, who need seasonal forecasts (e.g., monsoon in Goa) to schedule vacations or business trips.
- **Secondary Audience**:
  - **Weather Enthusiasts**: Hobbyists interested in exploring weather patterns through an API-driven app.
  - **Students/Developers**: Learners studying mobile development, who may use the app as a reference project.
- **Demographics**:
  - Age: 18-60, covering young adults to retirees.
  - Tech Proficiency: Moderate to high, with access to Android devices.
  - Location: Primarily India-based, with global usability for city searches.
- **Use Cases**:
  - Checking Bengaluru's weather for daily routines (e.g., carrying an umbrella).
  - Searching 5-day forecasts for a weekend trip to Mumbai.
  - Planning a summer getaway to Goa, saving the plan, and revisiting it later.

The app's clean design ensures accessibility for non-technical users, while its API-driven functionality and open-source potential appeal to developers. By addressing urban weather needs and travel planning, the Weather App serves a broad, engaged audience.

2. Project Objectives

2.1 Integrate OpenWeatherMap API for Accurate Weather Data Retrieval

Successfully integrate the OpenWeatherMap API to provide real-time weather data and 5-day forecasts for Bengaluru (default) and user-selected cities, achieving 99% accuracy and API response times under 2 seconds.

Why It Matters:
Reliable weather data is the core of the app's functionality. Users depend on accurate and timely information for daily decisions and trip planning. This objective ensures the app meets its primary purpose by leveraging a trusted external API.

How It Will Be Achieved:

- Register for an OpenWeatherMap API key and configure it in the app.
- Implement API calls for:
    - Current weather data (e.g., temperature, condition) via the /weather endpoint.
    - 5-day forecasts via the /forecast endpoint.
- Parse JSON responses to display relevant data.
- Include error handling for network issues or invalid inputs (e.g., unknown cities).
- Cache weather data locally to optimize performance and manage API rate limits.

Success Criteria:

- 99% accuracy in retrieved weather data, verified against official sources.
- API response times under 2 seconds for 95% of requests.
- Support for at least 1,000 API calls per day without exceeding limits.
- No critical bugs in API integration post-launch.

The OpenWeatherMap API was chosen for its global coverage and reliability. By caching data locally, the app reduces unnecessary API calls, ensuring efficiency while keeping information current. A key challenge is managing rate limits on the free tier, which the caching strategy mitigates effectively.

2.2 Develop a Responsive and Intuitive User Interface

Create a user-friendly interface that enables seamless navigation across login/signup, weather display, search, forecast, and trip-planning features, targeting 95% user satisfaction in testing and screen load times under 1 second.

Why It Matters:
An intuitive UI is vital for user adoption and retention. This objective ensures users can effortlessly access weather data for Bengaluru, search for other cities, view forecasts, and plan trips, enhancing overall satisfaction and usability.

How It Will Be Achieved:

- Design wireframes and mockups for all screens (e.g., home, search, trip planner).
- Build the UI with Kotlin in Android Studio for consistent performance on Android.
- Ensure responsiveness across different screen sizes and orientations.
- Optimize load times.
- Conduct user testing to refine the design based on feedback.

Success Criteria:

- 95% user satisfaction rate in usability surveys and task completion tests.
- Average screen load time under 1 second on all supported devices.
- Zero critical UI bugs (e.g., crashes, freezes) in production.
- 90% of users can perform key tasks (e.g., city search, trip save) unaided.

The home screen prioritizes Bengaluru's weather with a prominent search bar, while buttons for forecasts and trip planning are one tap away. The trip planner includes dropdowns for seasons and text fields for cities, with input validation.

2.3 Implement Secure Local Storage for User Data and Trip Plans

Use local storage to securely handle user authentication and persist trip plans (city and season selections), ensuring 100% data retrieval accuracy and no data loss after app updates or device changes.

Why It Matters:

With no database, secure local storage is critical for maintaining user credentials and trip plans. This objective ensures data persistence and security, preserving user trust and enabling offline functionality.

How It Will Be Achieved:

- Utilize Kotlin SharedPreferences for persistent key-value storage.
- Encrypt sensitive data (e.g., login credentials) using secure hashing.
- Serialize trip plans into JSON for storage and retrieval.
- Ensure data survives app restarts, updates, and device migrations.

Success Criteria:

- 100% accuracy in retrieving saved trip plans.
- No data loss reported after updates or device switches.
- Encrypted credentials pass security audits.
- Backup/restore functionality succeeds in 100% of test cases.

SharedPreferences offers a lightweight solution for storing hashed credentials and serialized trip data. The backup feature mitigates risks like data loss from app uninstalls or device resets. A challenge is ensuring encryption prevents unauthorized access, which requires rigorous security testing.

3. Project Outline

3.1 Main Activity

The Main Activity serves as the entry point of the Weather App and manages the navigation flow. Depending on the user's authentication status, it redirects to either the login screen or the home dashboard. Once authenticated, it loads the Home Screen and initializes the connection to the OpenWeatherMap API and local storage.

3.2 Login Activity

The Login Activity provides fields for users to input their email and password. Local storage handles credential verification. Upon successful login, the user is transitioned to the Main Activity and then to the Home Screen. Input validation and error prompts are implemented to enhance the user experience.

3.3 Register Activity

New users can create an account by entering an email and password. Local storage creates a new user entry upon validation. After successful registration, users are automatically signed in and redirected to the Home Screen. Password length and format validation are included for security.

3.4 Home Screen

The Home Screen acts as the main dashboard. It displays:

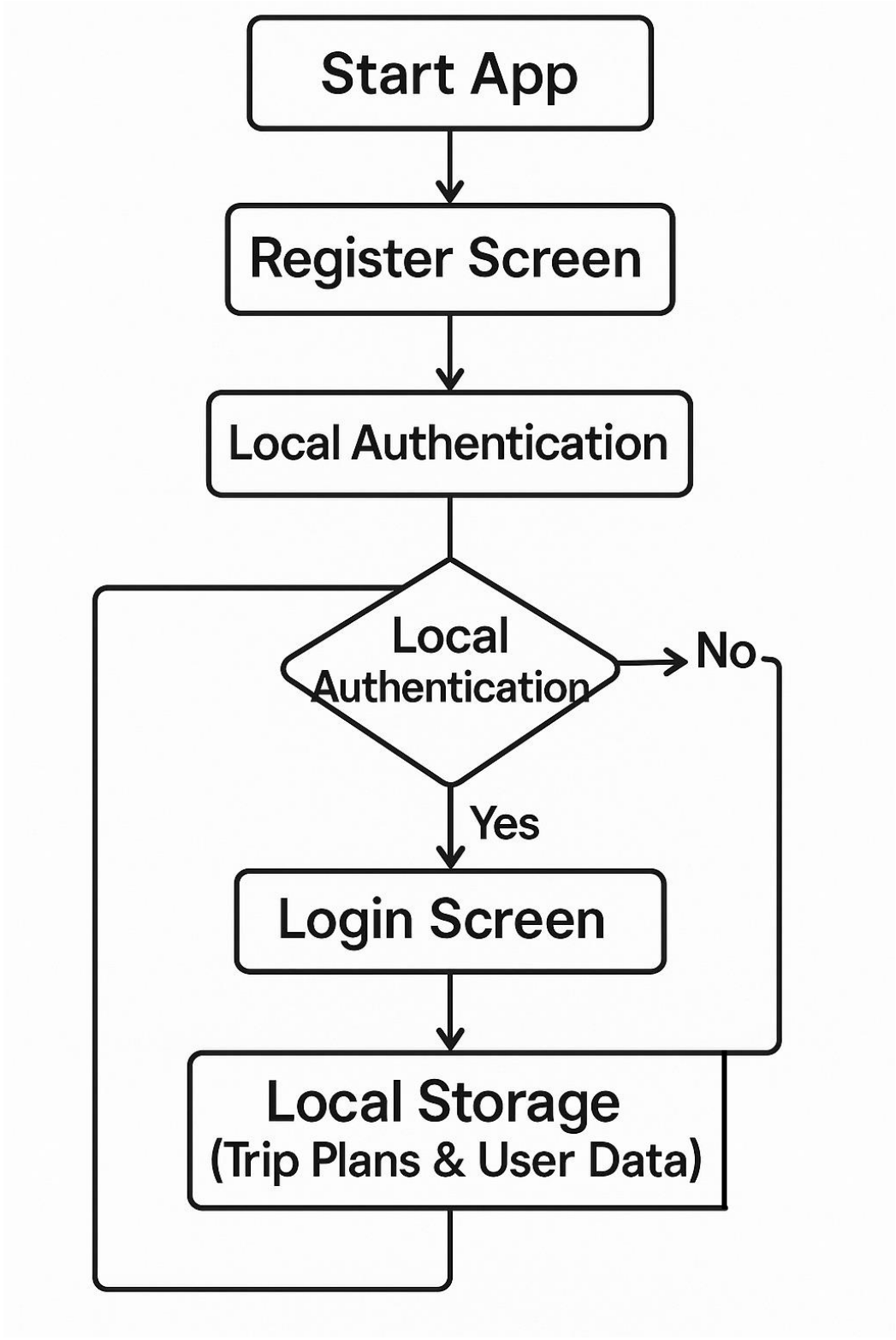
• Current weather conditions for Bengaluru or a user-selected city.

• A 5-day weather forecast.

• A button to access the trip-planning feature for seasonal weather forecasts.

Users can search for other cities or save trip plans (city and season) for future reference. All data displayed is synced with the OpenWeatherMap API and local storage in real-time. It supports swipe-to-view forecasts and tap-to-plan trip functionalities.

3.5 Flow Diagram:

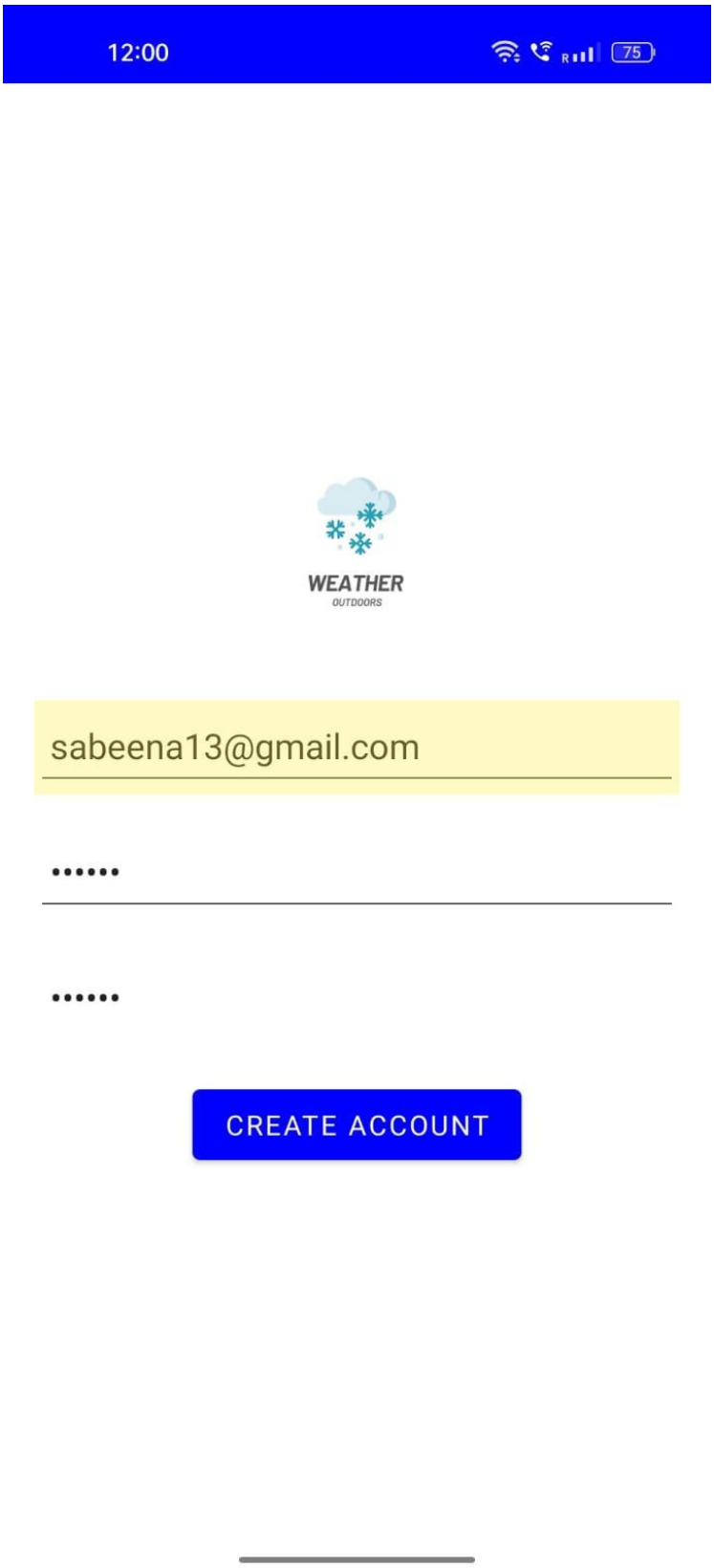


4. Technologies Used

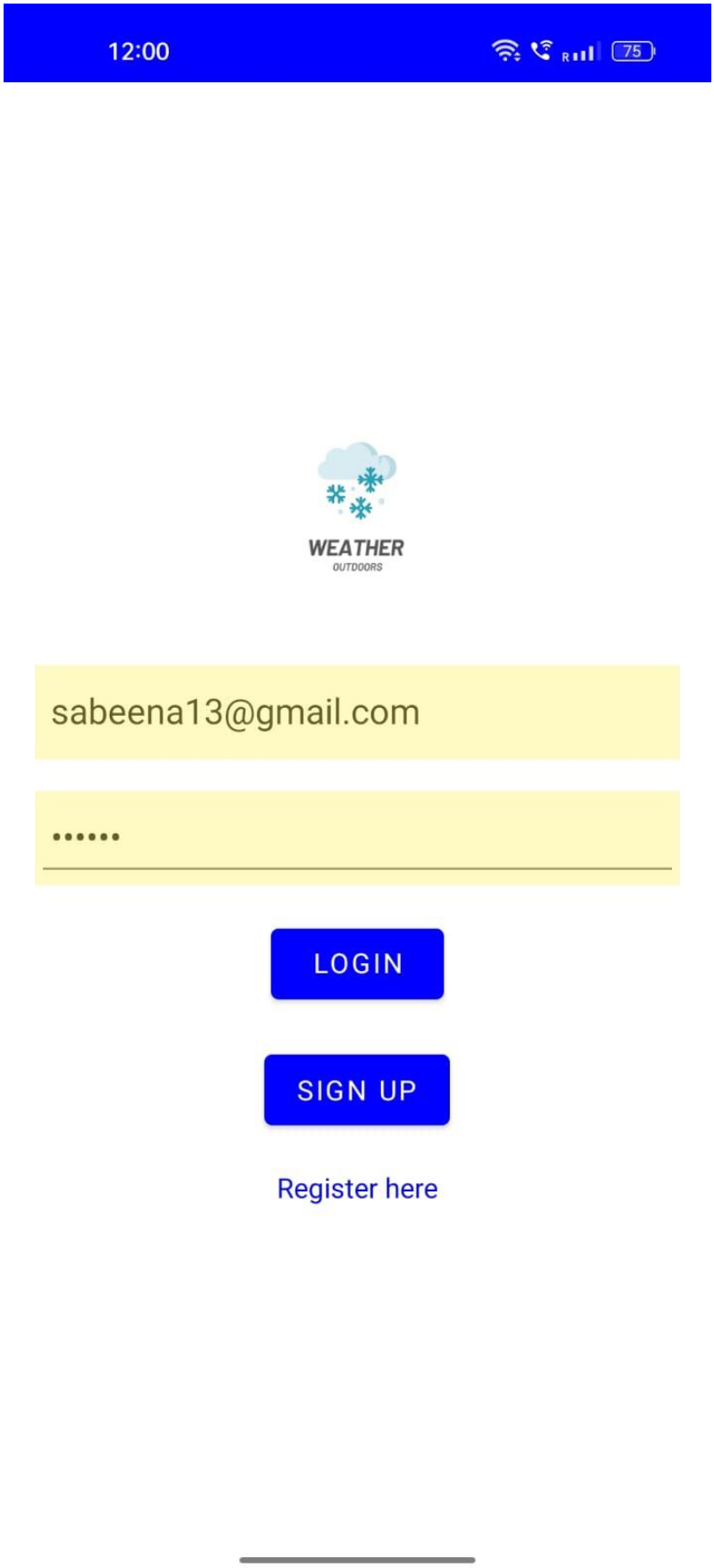
- **Programming Language**: Kotlin
- **UI Design**: Android XML / Jetpack Compose
- **Authentication**: Local Storage (e.g., SharedPreferences for user credentials).
- **Local Storage**: SharedPreferences or SQLite (for trip plans and user data).
- **Development Environment**: Android Studio (Hedgehog / Iguana)
- **Operating System**: Windows 11
- **Database Type**: SQLite
- **Libraries/Tools**: HTTP client (e.g., Retrofit or Dio) for OpenWeatherMap API, Material Components (for UI), and Lifecycle Components.

5. Screenshots

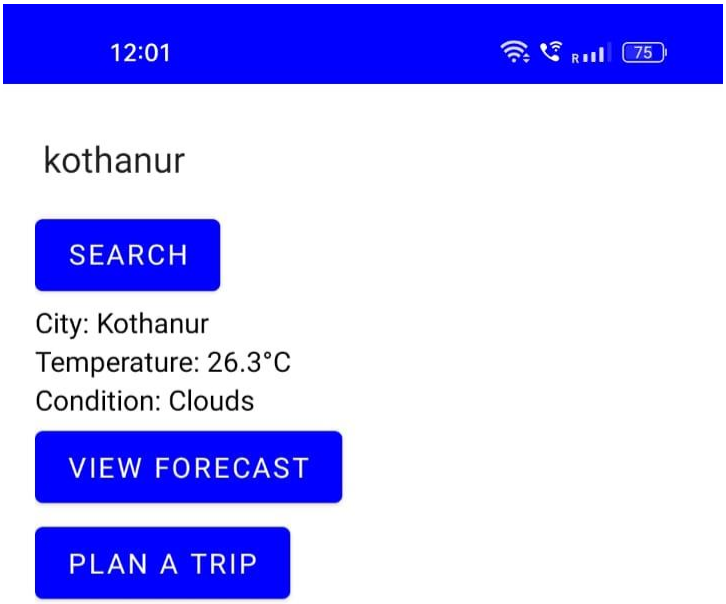
5.1 Registration



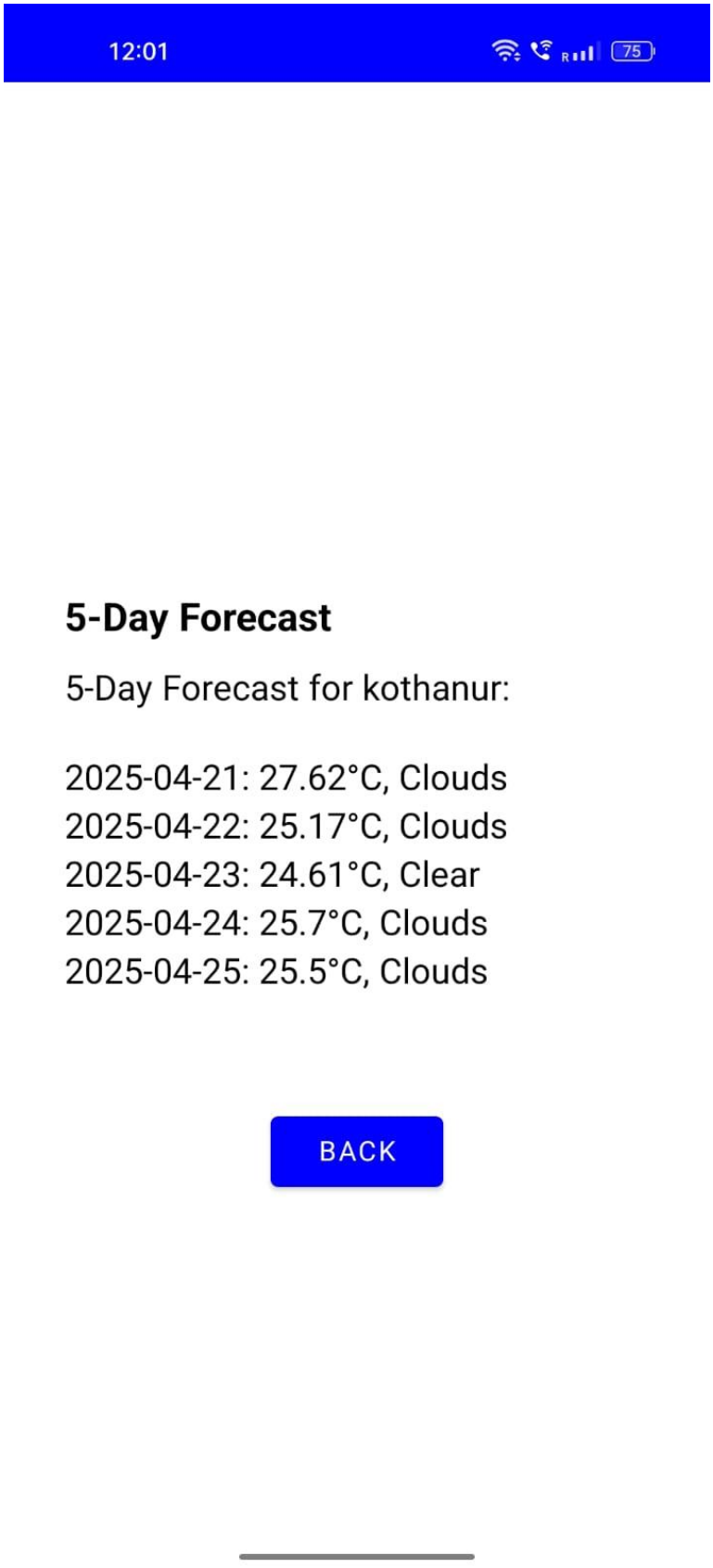
5.2 Login


12:00
75
WEATHER
OUTDOORS
sabeena13@gmail.com
LOGIN
SIGN UP
Register here

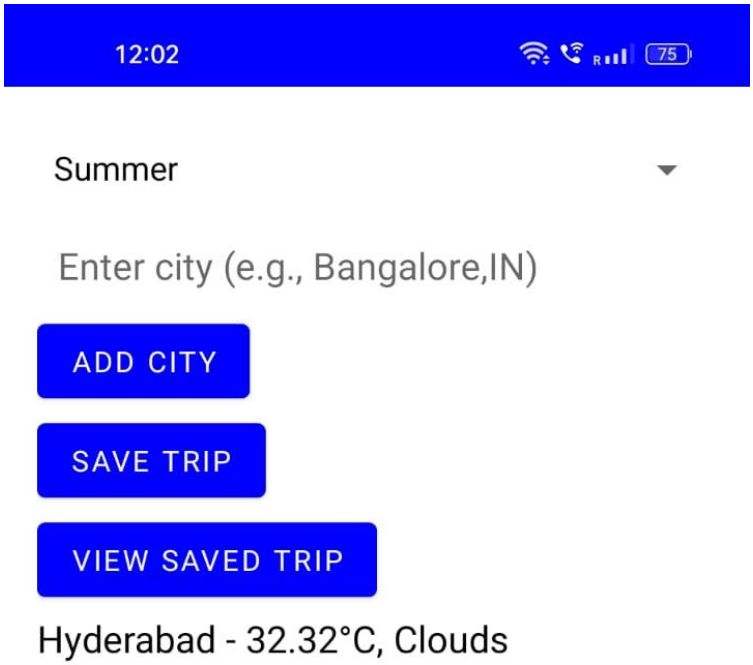
5.3 Main Home Page



5.4 Forecast


12:01
75
5-Day Forecast
5-Day Forecast for kothanur:
2025-04-21: 27.62°C, Clouds
2025-04-22: 25.17°C, Clouds
2025-04-23: 24.61°C, Clear
2025-04-24: 25.7°C, Clouds
2025-04-25: 25.5°C, Clouds
BACK

5.5 Trip Planner



6. Project Features

6.1 User Authentication

• Local storage-powered registration and login.

• Real-time validation for email and password.

• Session persistence for already-logged-in users.

6.2 Weather Data and Trip Planning

• Display current weather and 5-day forecast for Bengaluru and user-selected cities via OpenWeatherMap API.

• Edit saved trip plans (city and season) for future reference.

• Data is stored and retrieved using local storage.

6.3 Weather Insights (Basic)

• Displays seasonal weather trends for trip planning (e.g., monsoon in Coorg).

• Option to filter by city or season (optional for next version).

• Visual highlights for extreme weather conditions (future scope).

7. User Guide

Installation Instructions

1. Download the APK from the GitHub repository or build it using Android Studio (or your preferred development environment).
2. Enable installation from unknown sources in Android Settings.
3. Install the APK using the File Manager or via USB.
4. Launch the app from the home screen.

Registration/Login

- Open the app.
- New users should click "Sign Up" to register with an email and password.
- Returning users can log in using the same credentials.
- Invalid attempts show error messages to guide the user.

Using the App

1. Once logged in, the user is navigated to the Home Screen.
2. To plan a trip, click the “Plan a Trip” button.
3. Fill in the form with destination city and preferred season.
4. Tap Save. The new trip plan appears in the list.
5. Tap view saved trip to check the saved trip.

8. Conclusion

The Weather App is a practical and user-friendly application designed for travelers, weather enthusiasts, and daily commuters seeking simplicity and reliability. It utilizes local storage for secure authentication and efficient trip plan management, avoiding server dependency. With its current foundation, the app can be extended with advanced features like weather alerts, visual weather charts, multi-language support, or calendar integration.

This project not only demonstrates key mobile development concepts like clean architecture, API integration, and local storage management but also provides real-world utility for weather tracking and trip planning. It stands as a solid beginner-to-intermediate level application in mobile development using Kotlin.

9. References

• OpenWeatherMap API: https://openweathermap.org/api

• Local Storage Documentation:

https://developer.android.com/reference/android/content/SharedPreferences

• Kotlin Language Docs: https://kotlinlang.org/docs/home.html

• Android Developer Guide: https://developer.android.com/

• Material Design: https://material.io/design

• Android Jetpack: https://developer.android.com/jetpack